Imperial College London

BSc/MSci EXAMINATION May 2017

This paper is also taken for the relevant Examination for the Associateship

PRINCIPLES OF INSTRUMENTATION

For 3rd/4th-Year Physics Students

Thursday, 25th May 2017: 10:00 to 12:00

The paper consists of two sections: A & B.
Section A contains one question, comprising of small parts. [40 marks total]
Section B contains four questions on selected parts of the course. [30 marks each]
Table of Laplace transformations is included at the end of the paper.
Candidates are required to:
Answer **ALL** *parts of Section A and* **TWO** *questions from Section B.*
Marks shown on this paper are indicative of those the Examiners anticipate assigning.

General Instructions

Complete the front cover of each of the THREE answer books provided.

If an electronic calculator is used, write its serial number at the top of the front cover of each answer book.

USE ONE ANSWER BOOK FOR EACH QUESTION.

Enter the number of each question attempted in the box on the front cover of its corresponding answer book.

Hand in THREE answer books even if they have not all been used.

You are reminded that Examiners attach great importance to legibility, accuracy and clarity of expression.

Go to the next page for questions

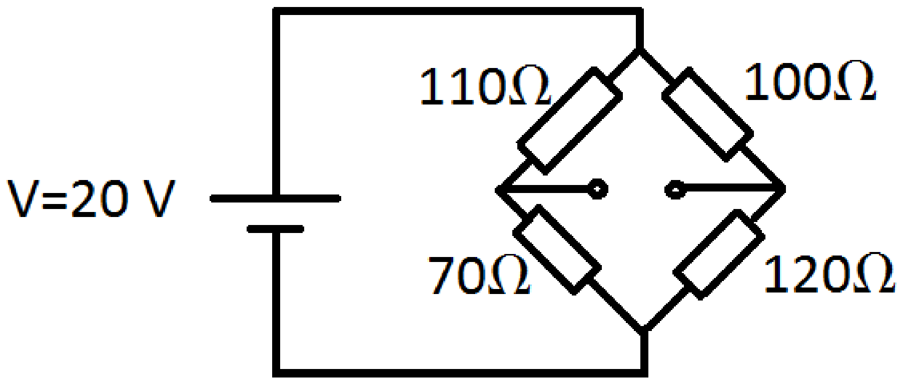
SECTION A

1. (i) (a) Briefly state Thévenin's theorem.

(b) Sketch the Thévenin equivalent circuit for the Wheatstone bridge shown in the diagram below and calculate the Thévenin voltage.

(c) Calculate the Thévenin resistance.

[11 marks]



(ii) (a) Briefly state Norton's theorem.

(b) Sketch the Norton equivalent circuit for the Wheatstone bridge shown in the diagram above and calculate the Norton current.

(c) Give the Norton resistance.

[9 marks]

(iii) (a) Briefly state the properties of the ideal operational amplifier and give the two 'golden-rules' used to simplify the analysis of circuits containing them.

(b) The diagram below shows an op-amp which is assumed to be ideal. Show that

$$V_{out} = -\frac{1}{RC}\int V_{in}\mathrm{d}t$$

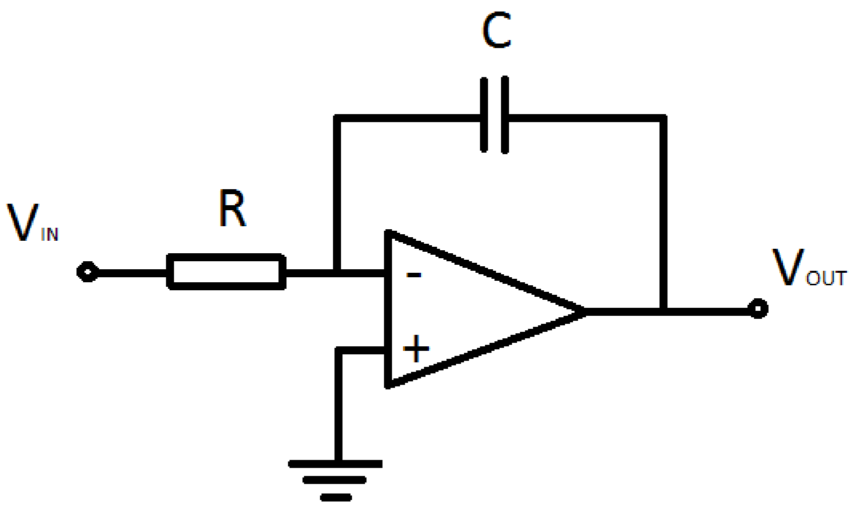
(c) With $R = 20\,\mathrm{k}\Omega$ and $C = 50\,\mu\mathrm{F}$, sketch V_{out} for

$$V_{in}(t) = \begin{cases} 0 & t < -a \\ t & -a \leq t \leq a \\ 0 & t > a \end{cases}$$

(d) Briefly state the properties of typical 'real-world' operational amplifiers.

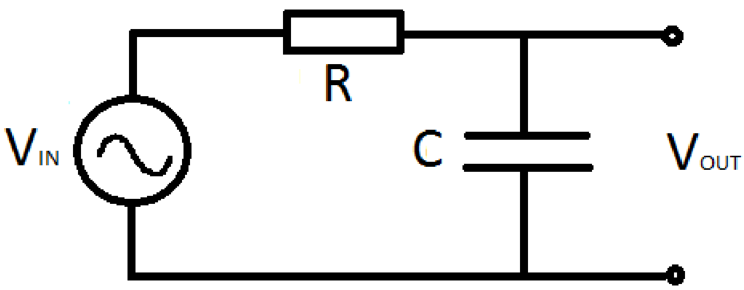
[20 marks]

[Total 40 marks]



SECTION B

2. The RC circuit in the diagram below has an AC voltage signal applied to it.



(i) (a) Give an expression for the total impedance of the circuit, including its frequency dependence.

(b) Write an equation for the current flowing out of the voltage source.

[5 marks]

(ii) (a) Define the voltage gain G and show that its absolute value can be expressed as

$$|G| = \frac{1}{\sqrt{\omega^2 R^2 C^2 + 1}},$$

where ω is the angular frequency of the AC source.

(b) Calculate the frequency dependent phase angle between the input and output voltages.

[8 marks]

[This question continues on the next page . . .]

(iii) (a) Assuming R=2 kΩ and C=0.5 μF calculate the cut-off frequency of the circuit and make a sketch of the Bode plot.

(b) Explain the behaviour of this circuit as a filter and propose methods to improve its behaviour in terms of steepness of the frequency roll-off, flatness of the pass-band and sharpness of the knee.

[8 marks]

(iv) (a) Explain briefly properties of the thermal noise in electronic circuits and methods of its reduction.

(b) Consider an instrument, operation of which is dominated by thermal noise at 300 K. Calculate the operational temperature required to improve the signal to noise ratio by 1 dB.

[9 marks]

[Total 30 marks]

3. (i) (a) Define the Laplace transform $F(s)$ of the continuous function $f(t)$.

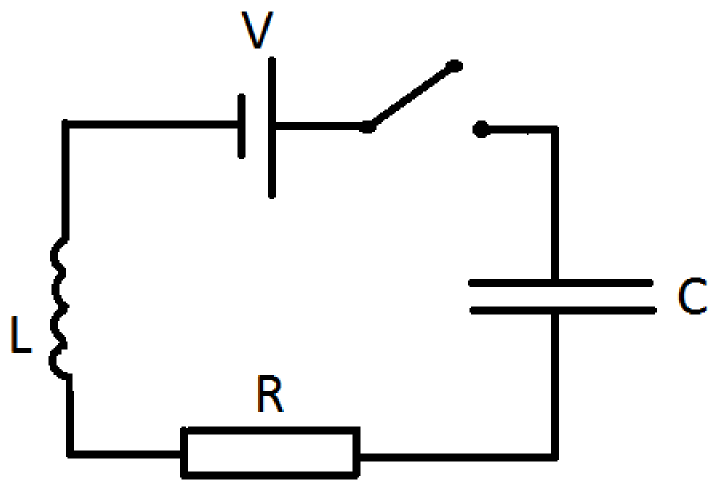
(b) Prove the transform's first derivative property:

$$\mathcal{L}\{\frac{df(t)}{dt}\} = sF(s) - f(0+)$$

where $f(0+)$ is the initial condition of the system.

[7 marks]

(ii) Consider the LRC circuit shown below consisting of a DC voltage source of 200 V, inductance of 1 H, capacitance of 0.04 F, resistance of 8 Ω and a switch. Assume that when the switch is turned on at time $t = 0$, the capacitor is charged and its initial voltage is 100 V.



(a) Write the differential equation describing the current flowing in the circuit.

(b) Write the differential equation describing the evolution of charge on the capacitor.

(c) Take the Laplace transform of the equation describing the charge on the capacitor and write its corresponding s-domain representation.

(d) By performing the inverse Laplace transform, find an expression for the charge stored on the capacitor for time $t > 0$ after closing the switch.

(e) Calculate current flowing in the circuit $I(t)$ after closing the switch and sketch its behaviour.

[23 marks]

[Total 30 marks]

4. (i) A photomultiplier tube (PMT) with 11 dynodes has an electron gain of 10^7.

(a) Calculate how many additional electrons are being released, on average, at each dynode.

(b) Assuming that this PMT has a photocathode with quantum efficiency of 20% for light with 400 nm wavelength, calculate the average total charge created per individual photon of that wavelength entering the tube.

[9 marks]

(ii) The PMT is connected to the data acquisition electronics via a co-axial cable with a capacitance per metre $C' = 82\,\mathrm{pF\,m^{-1}}$ and inductance per metre $L' = 0.205\,\mu\mathrm{H\,m^{-1}}$.

(a) Calculate the characteristic impedance of the cable and explain its physical meaning.

(b) Calculate the time delay caused by the cable of length 70 m.

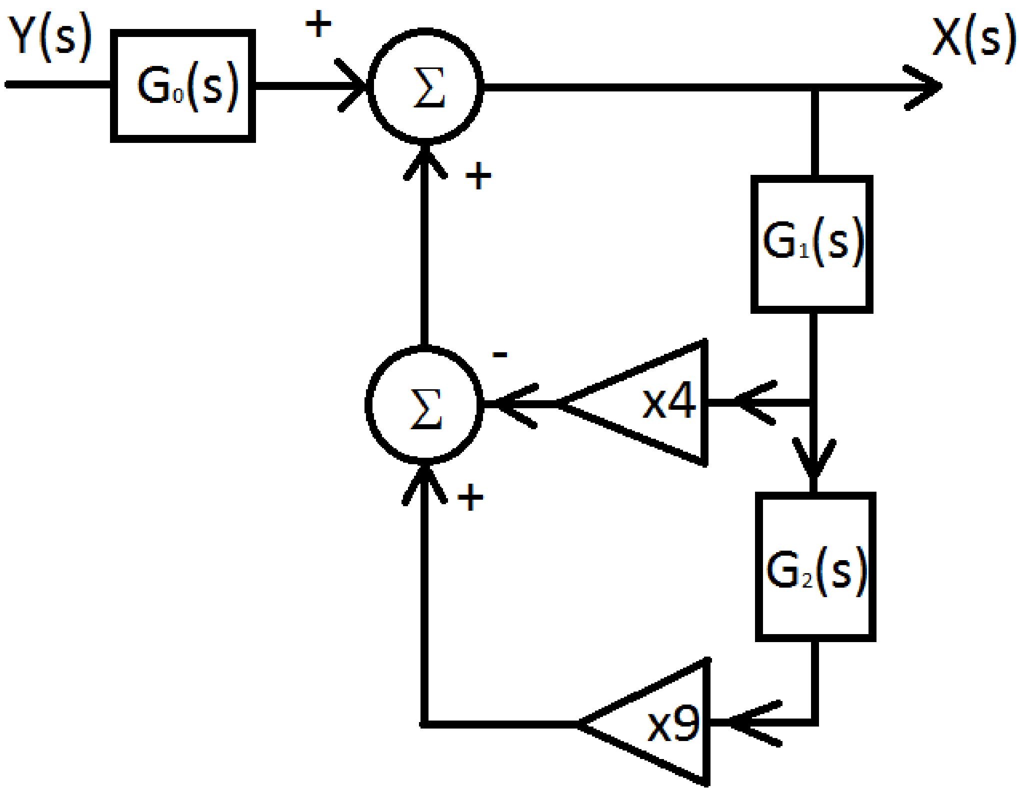
(c) Estimate the voltage on the cable created by a single photoelectron, assuming that average electron transit time spread through the PMT is 5.5 ns (Full Width at Half Maximum - FWHM).

[16 marks]

(iii) Define the so-called 'Dark Current' explaining its physical origin and its effect on measurements using PMTs. [5 marks]

[Total 30 marks]

5. The system shown below contains three functional blocks, two ideal amplifiers of gain 4 and 9, and contains both positive and negative feedback.



(i) (a) Calculate the system transfer function in terms of $G_0(s)$, $G_1(s)$ and $G_2(s)$.

(b) Assuming $G_0(s)$ has a time-domain impulse response of $16te^{-2t}$ find its s-domain representation.

(c) Assuming $G_1(s)$ is the ideal differentiator with unit gain find its s-domain representation.

(d) Sketch the circuit, which would have the transfer function $G_1(s)$ assuming it contains two ideal operational amplifiers.

(e) Assuming $G_2(s)$ is the ideal integrator with unit gain find its s-domain representation.

(f) Find the overall transfer function in terms of s.

[18 marks]

(ii) (a) Is the system stable? Explain your answer.

(b) Would it oscillate?

(c) Determine the system response to the unit step input in the time-domain.

[12 marks]

[Total 30 marks]

Table of Laplace Transforms

$f(t)$	$F(s) = \mathcal{L}\{f(t)\}$
$\delta(t)$	1
$u(t)$	$\frac{1}{s}$
t^n for $n > 0$	$\frac{n!}{s^{(n+1)}}$
e^{-at}	$\frac{1}{s+a}$
te^{-at}	$\frac{1}{(s+a)^2}$
$\frac{1}{b-a}(e^{-at} - e^{-bt})$	$\frac{1}{(s+a)(s+b)}$
$\sin(\omega t)$	$\frac{\omega}{s^2+\omega^2}$
$\cos(\omega t)$	$\frac{s}{s^2+\omega^2}$
$e^{-at}\sin(\omega t)$	$\frac{\omega}{(s+a)^2+\omega^2}$
$e^{-at}\cos(\omega t)$	$\frac{s+a}{(s+a)^2+\omega^2}$